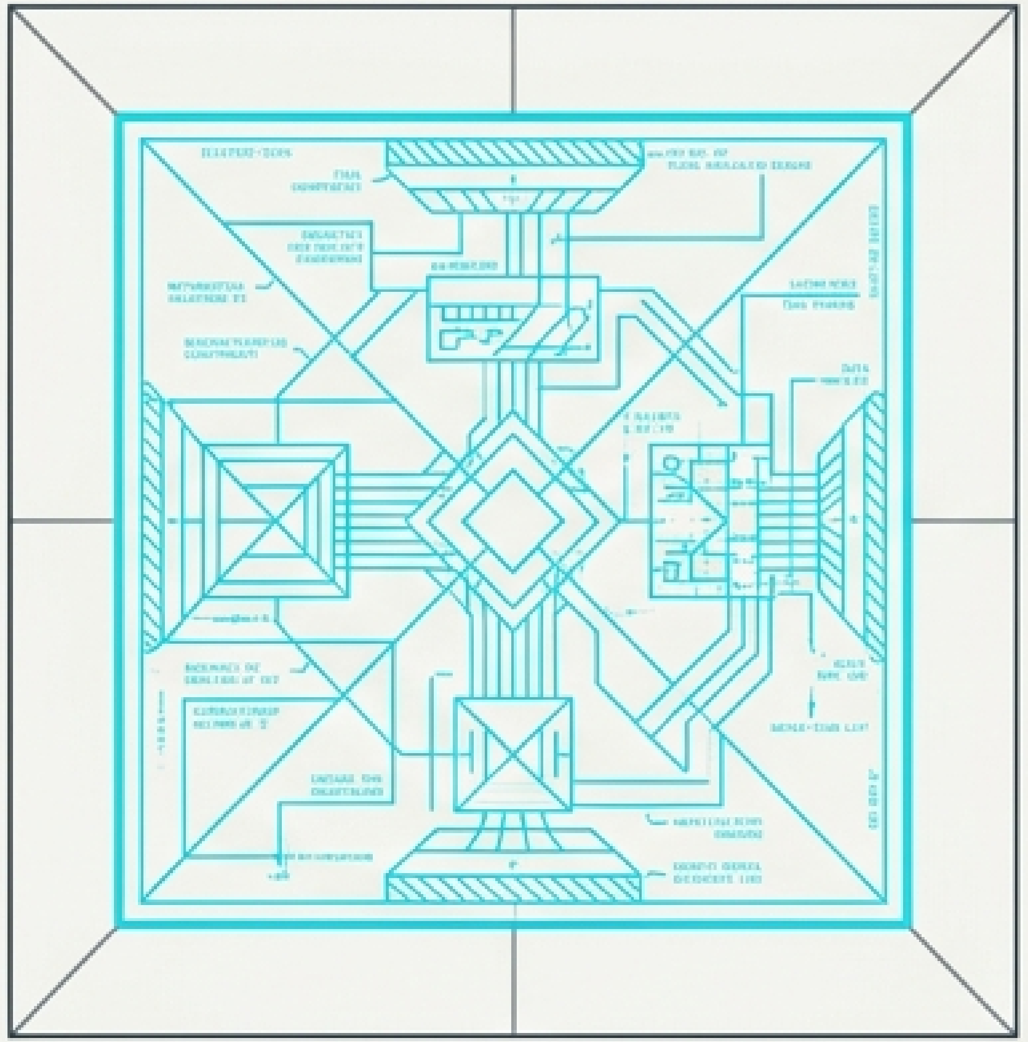
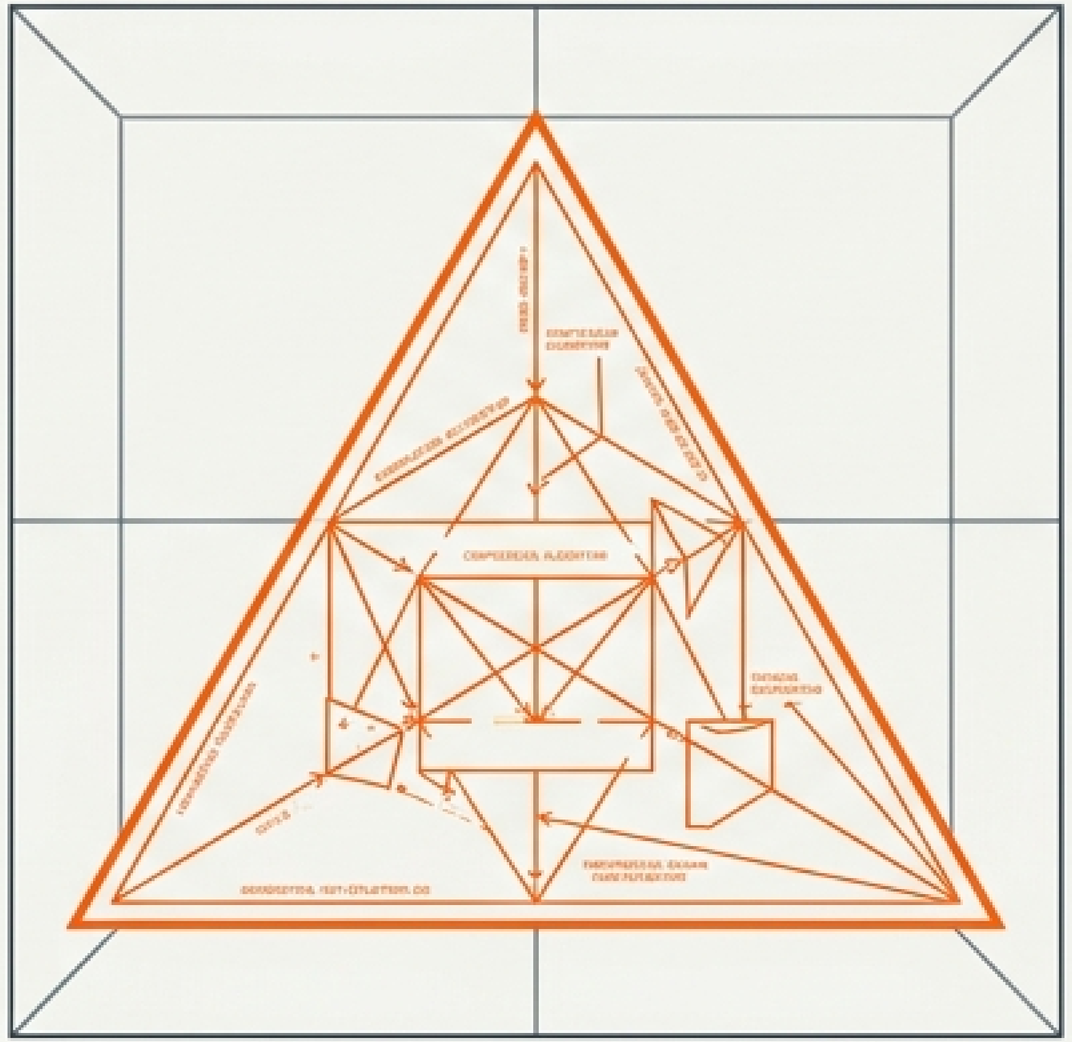
Against the Flattening Move in Systems Theory

Interdisciplinary synthesis often fails by equating distinct computational mechanisms under a single shared intuition.

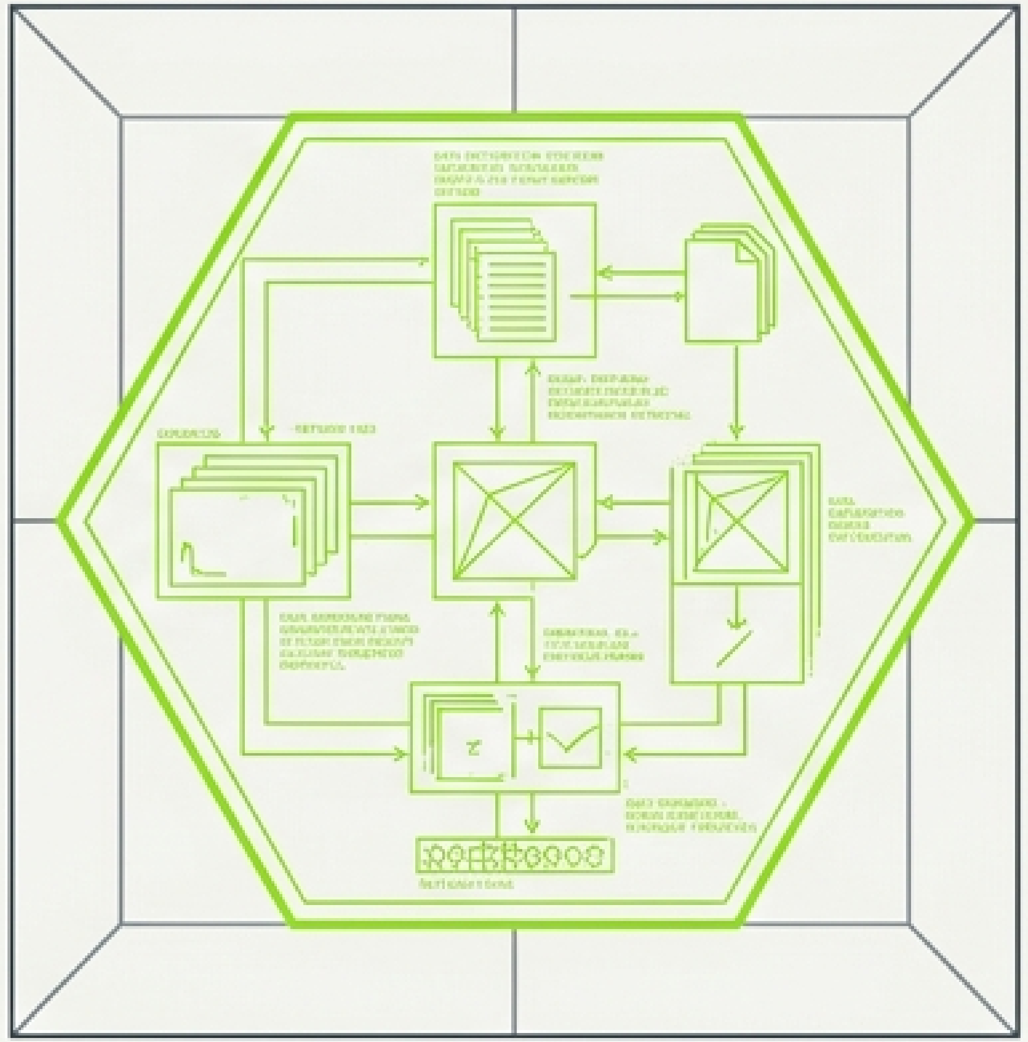
The Project: Three formal separations derived from Flyxion's July 2026 monographs.



Decoupling Boundaries



Decoupling Compression from Understanding

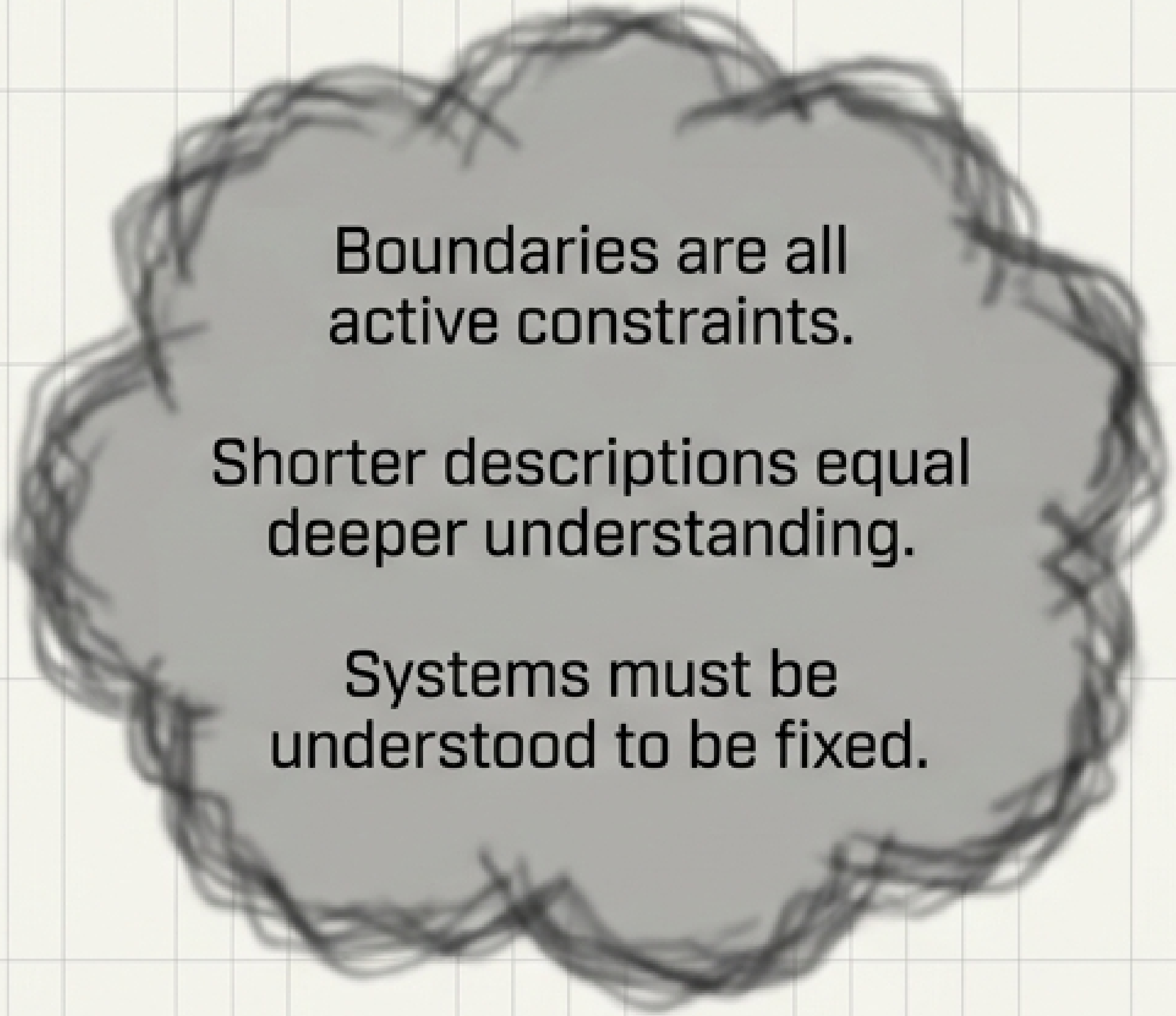


Decoupling Restoration from Explanation

THE DANGER OF INTERDISCIPLINARY METAPHORS

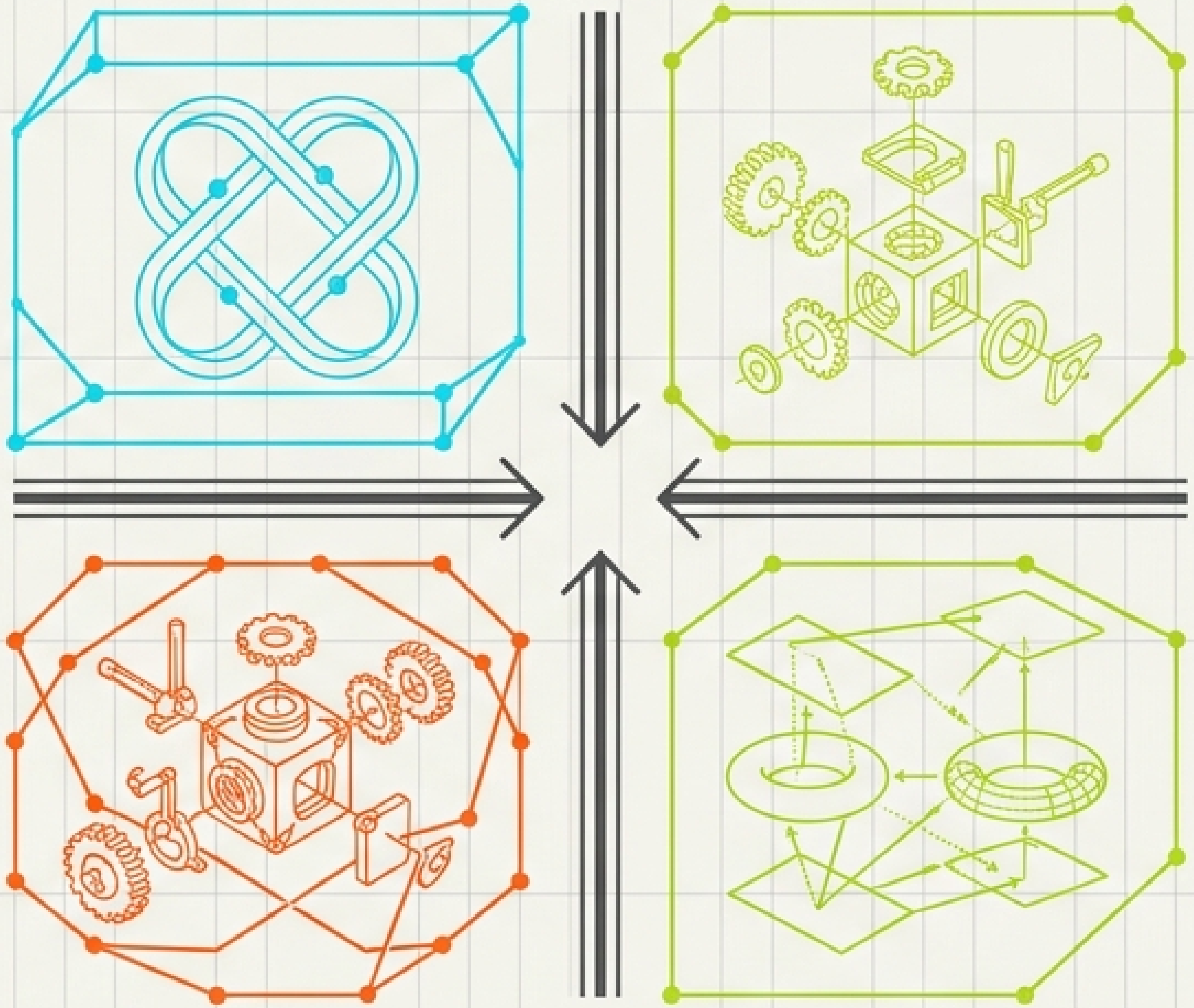
THE FLATTENING MOVE

Mistaking a shared intuition for a shared mechanism.



FORMAL TOPOLOGY

Moving from passive metaphors to structurally irreconcilable mechanisms.

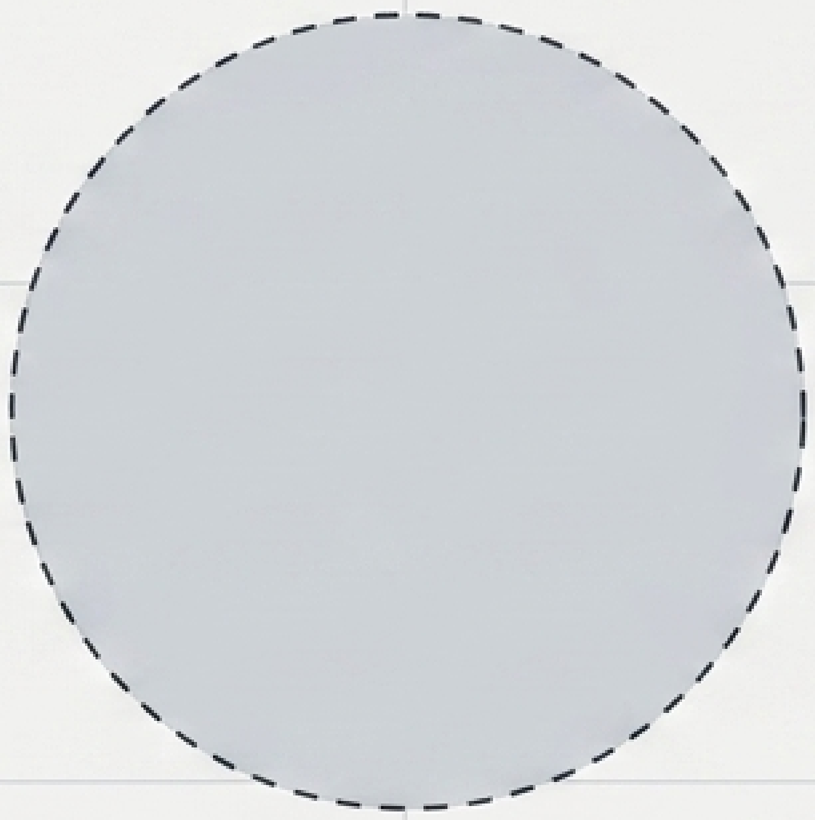


The failure of the passive picture is the only thing these theories actually share.

BOUNDARIES ARE COMPUTATIONAL OBJECTS, NOT PASSIVE GEOMETRY

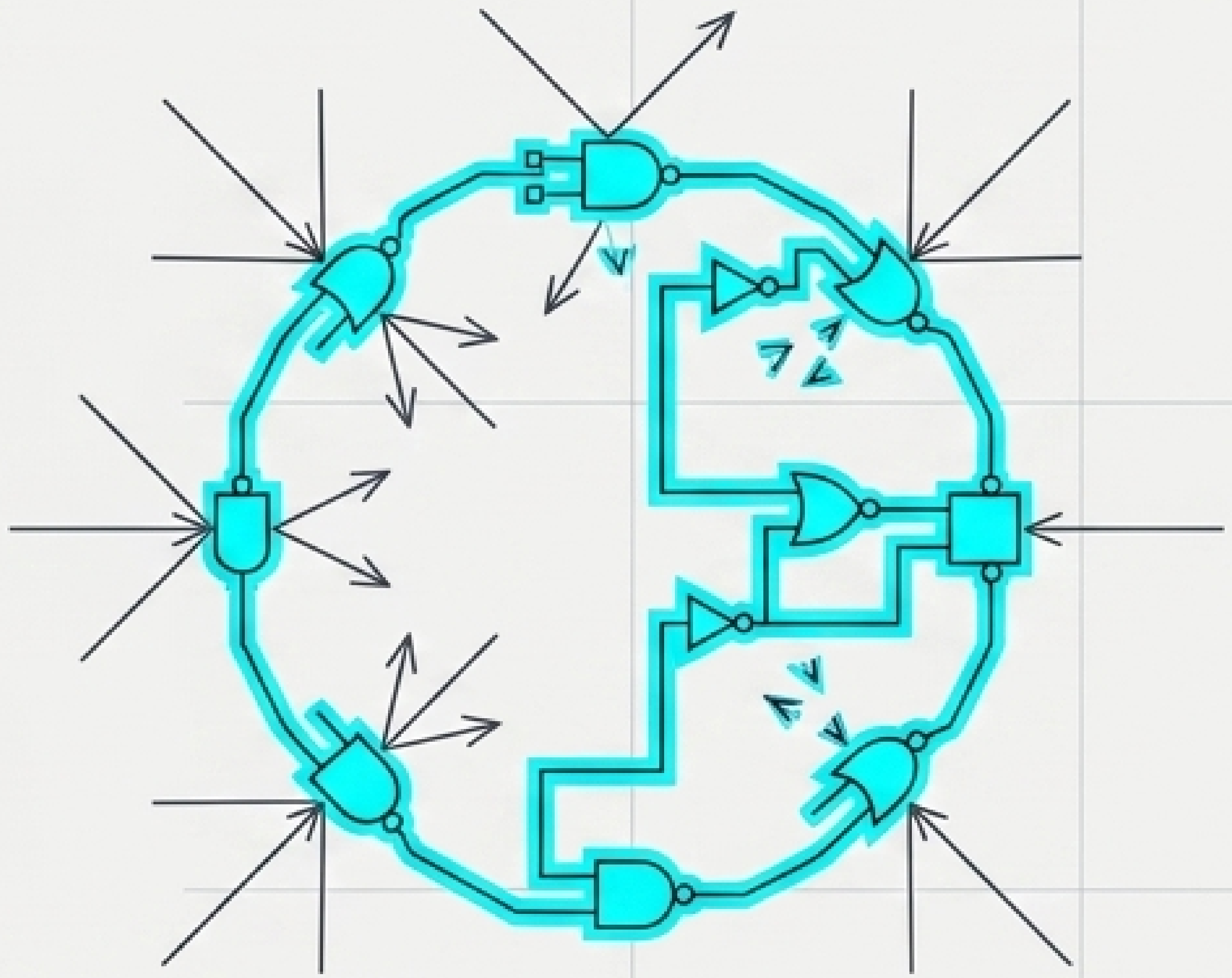
THE EVERYDAY ILLUSION

A boundary is a geometric or material fact prior to anything the boundary might do.



THE CONVERGENCE

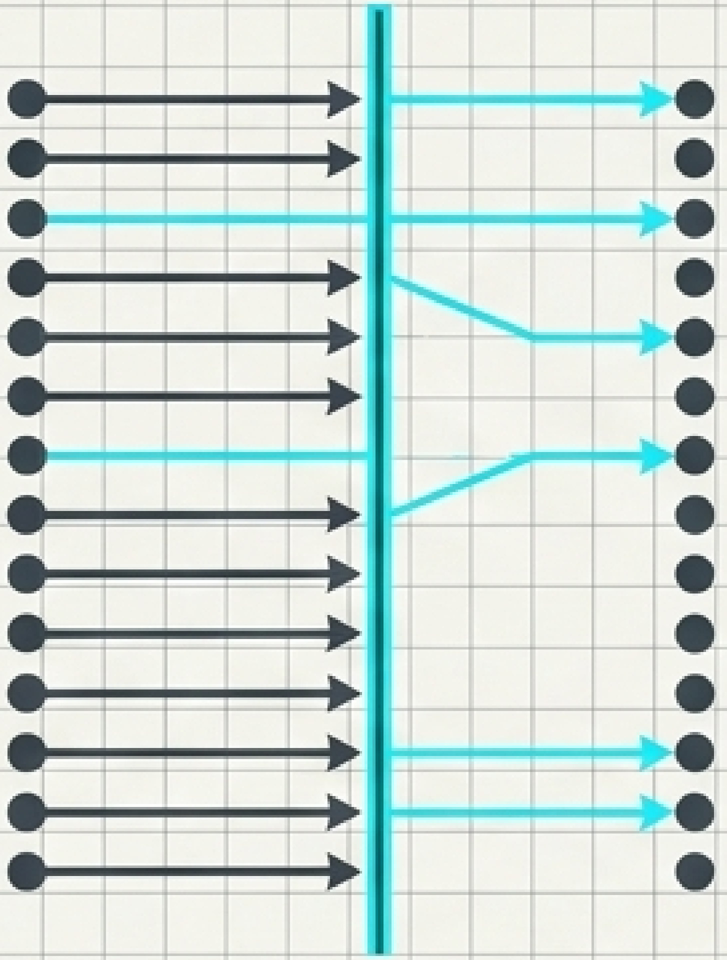
Three independent research programs conclude boundaries actively determine what a system is permitted to become.



The Flattening Trap: Naming the shared intuition ('boundaries as active constraints') is not the work. The mechanisms are fundamentally different.

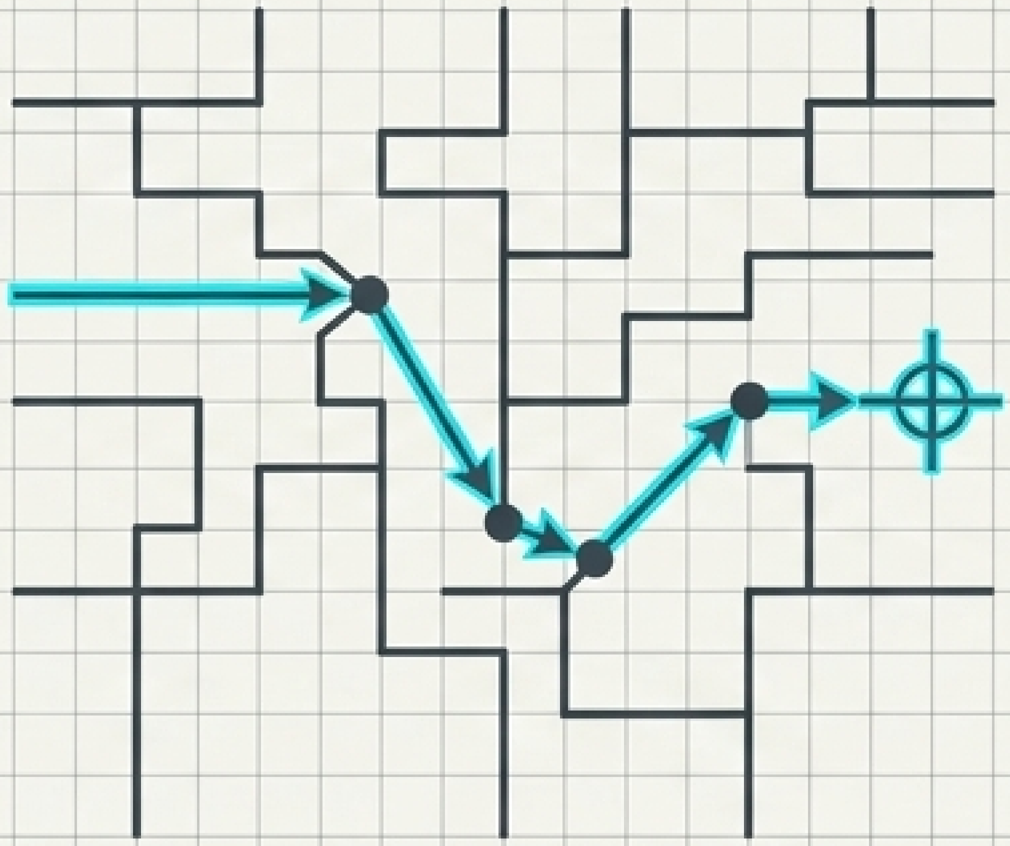
THREE DIVERGENT MECHANISMS OF BOUNDARY FORMATION

SCREENING
(Friston)



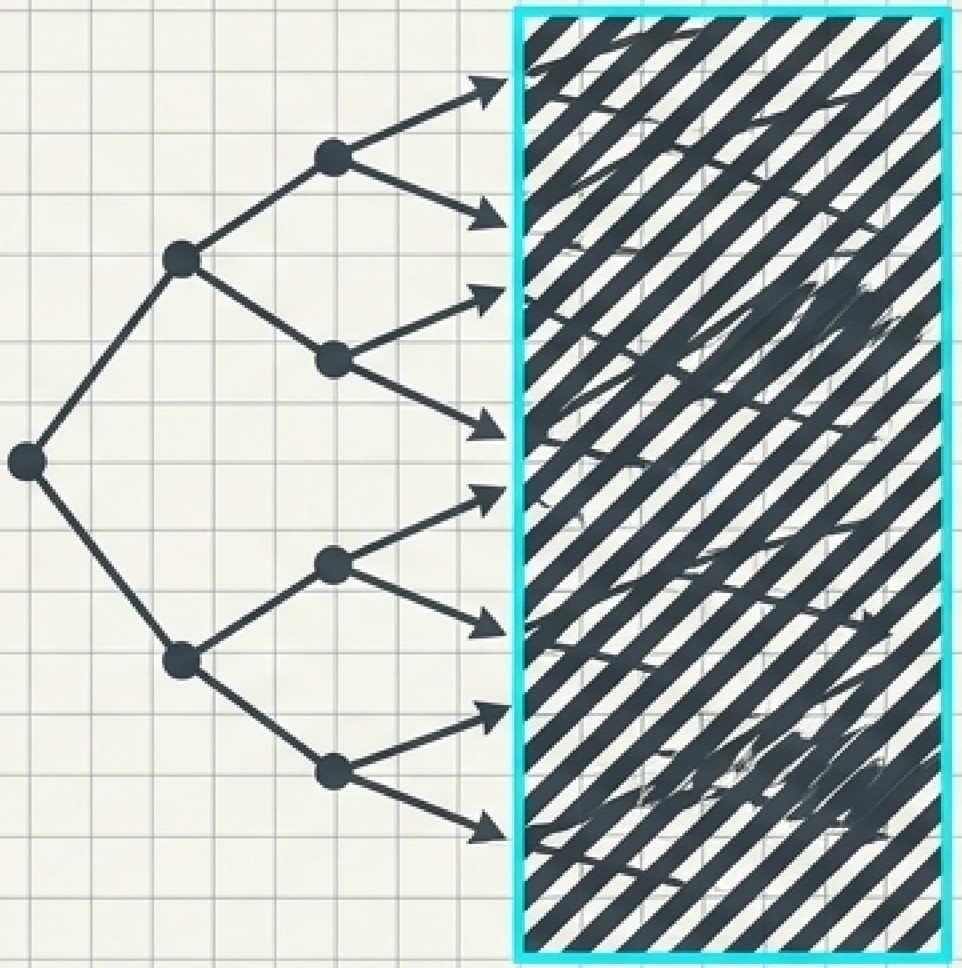
Definition: A Markov blanket inducing conditional independence between internal and external states.

PURSUIT
(Levin)



Definition: A bioelectric memory layer storing a target morphology and driving correction toward it.

DELETION
(RSVP/Flyxion)

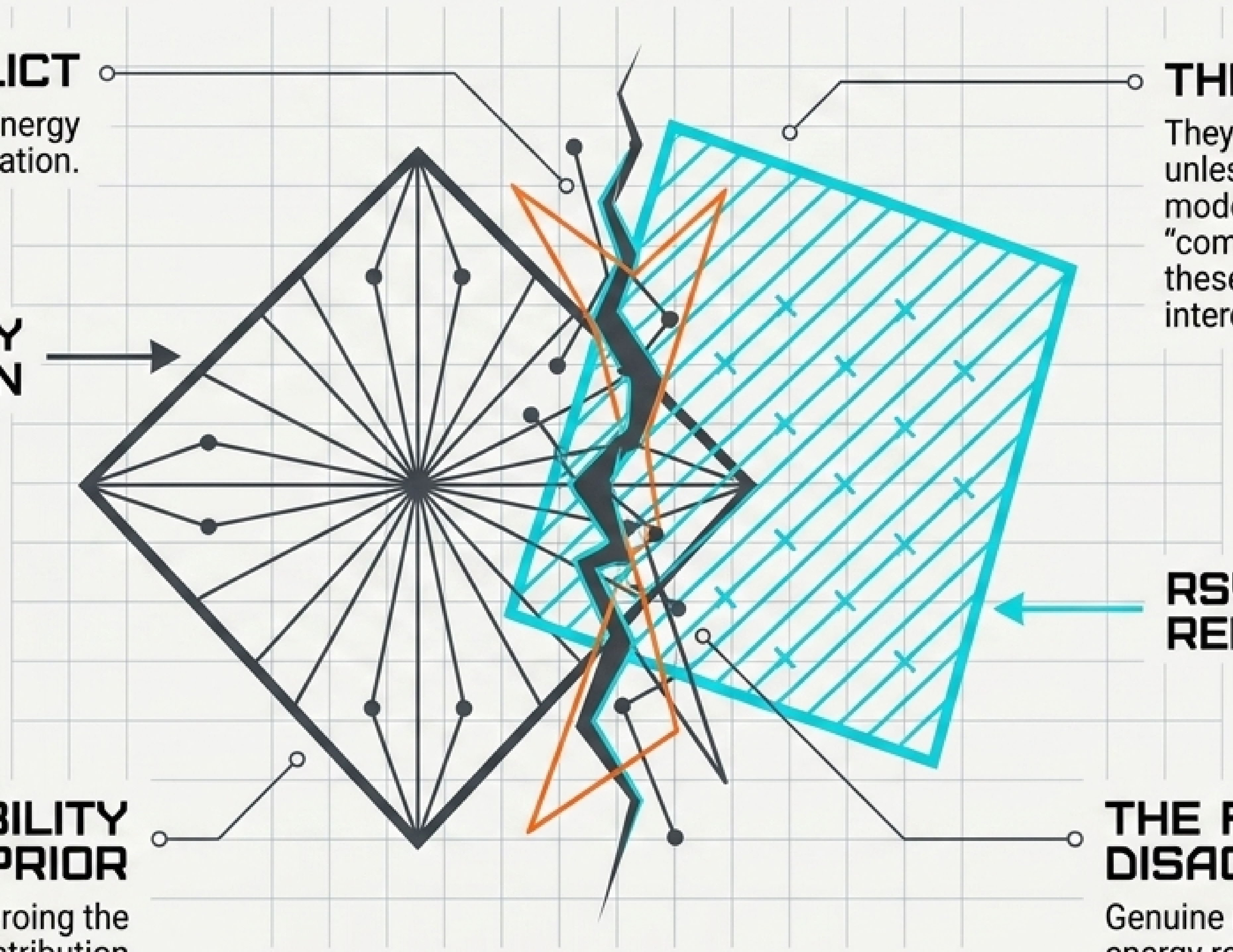


Definition: An irreversible commitment that permanently deletes a region from the system's space of possible futures.

THE BOUNDARY MATRIX: IRRECONCILABLE OPERATIONS

	FRISTON (SCREENING)	LEVIN (PURSUIT)	RSVP (DELETION)
MECHANISM	Absence of causal edges.	Stored target state.	Elimination of trajectory-space.
EVALUATED BY	Synchronic joint state (current moment).	Distance to a target state (future goal).	Diachronic history (accumulated past).
PRESUPPOSES	A joint distribution.	A stored pattern.	An accumulating record of removal.

THE POINT OF FRICTION: WHERE THE THEORIES ACTIVELY COLLIDE



THE CONFLICT

RSVP Refusal vs. Free-Energy Minimization.

FREE-ENERGY MINIMIZATION

THE INADMISSIBILITY PRIOR

RSVP formalizes refusal as zeroing the support of a trajectory distribution (deletion), not merely reweighting it.

THE RESOLUTION

They are mathematically incompatible unless the active-inference generative model is extended with latent “commitment variables”—proving these are distinct categories, not interchangeable metaphors.

RSVP REFUSAL

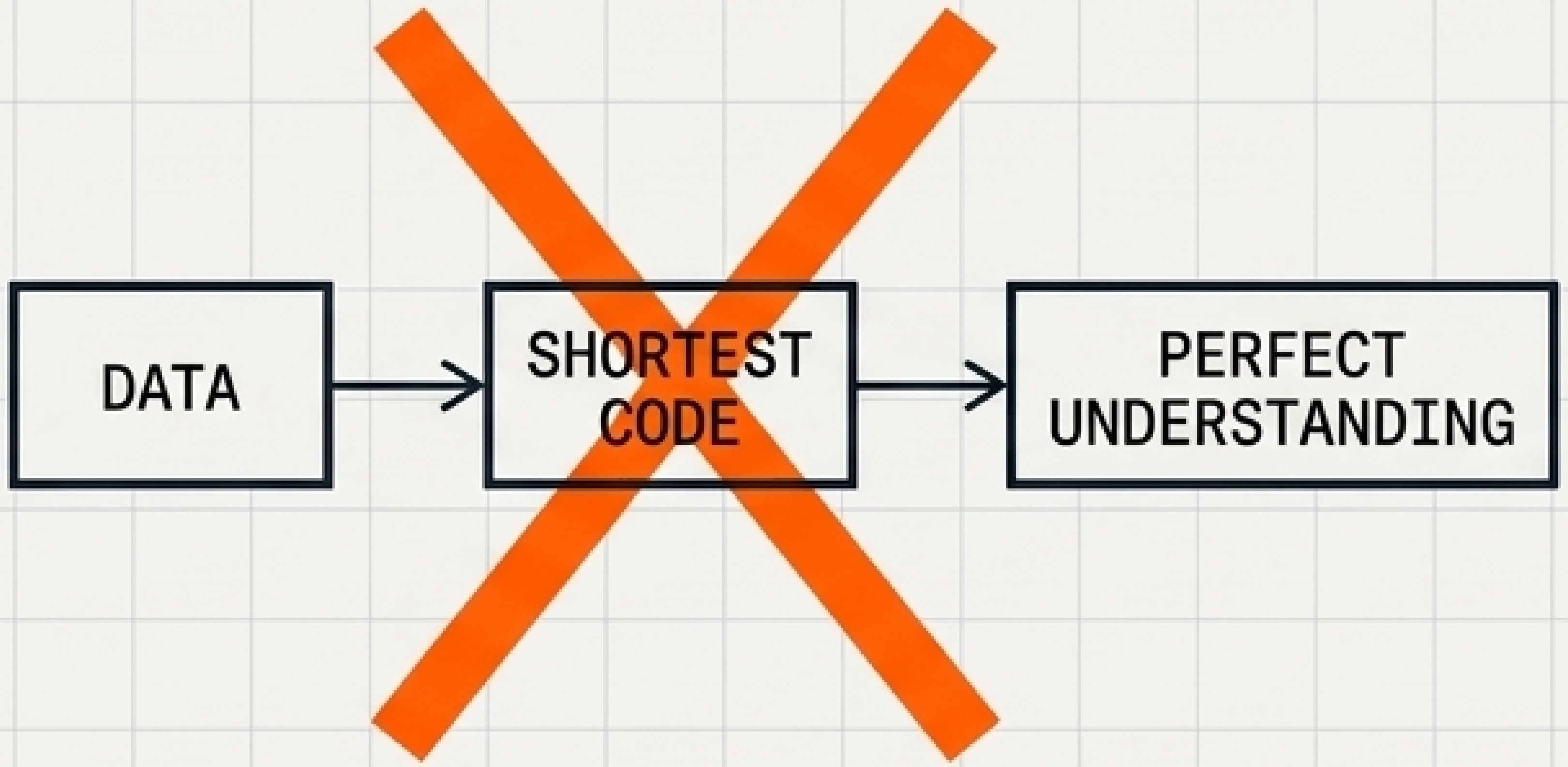
THE FALSIFIABLE DISAGREEMENT

Genuine refusal increases expected free energy relative to a standard generative model by removing a low-surprise path.

THE COMPRESSION FALLACY IN REPRESENTATIONAL CAPACITY

THE ACCEPTED HEURISTIC

The shortest description of a pattern is the correct measure of how well that pattern is understood (Schmidhuber, minimum-description-length principle).



THE FORMAL REALITY: An ontological triple $(X, \sim, T)$ requires distinguishing between what a language can express in principle and what it can reach in practice.

TWO DISTINCT VARIABLES:

1. **Coding Deficit ($\delta_T(x)$):** Excess length over Kolmogorov-optimal description.
2. **Distinguishability Deficit ($\Delta_T(x)$):** The gap in capacity to separate observationally distinct objects.

THE DEFICIT PROXY ASYMMETRY FUNNEL

The Proof

Perfect compression forces perfect distinguishability.

The Fallacy

The converse is unproven.

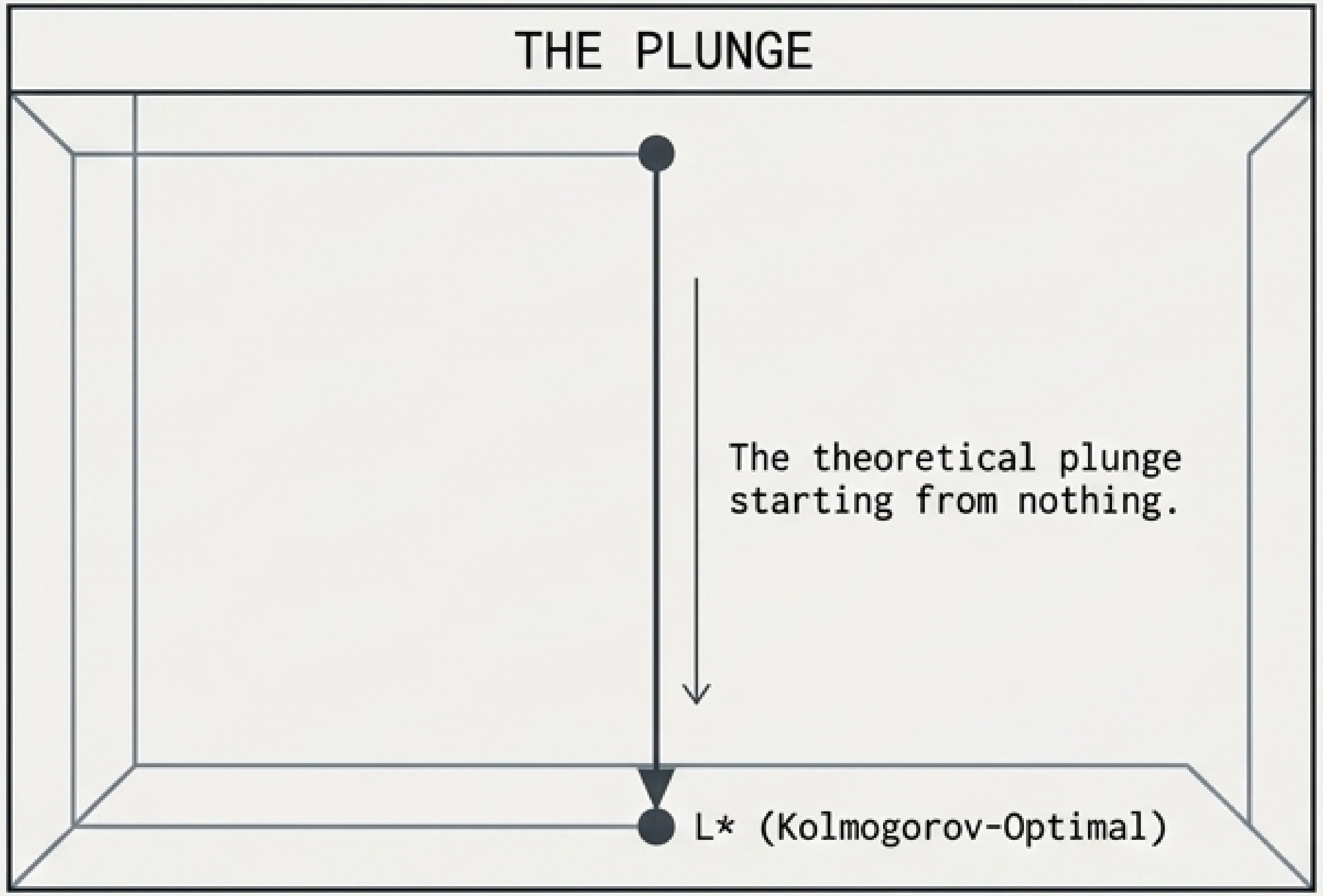
A system can carry a large coding deficit while carrying almost zero distinguishability deficit.

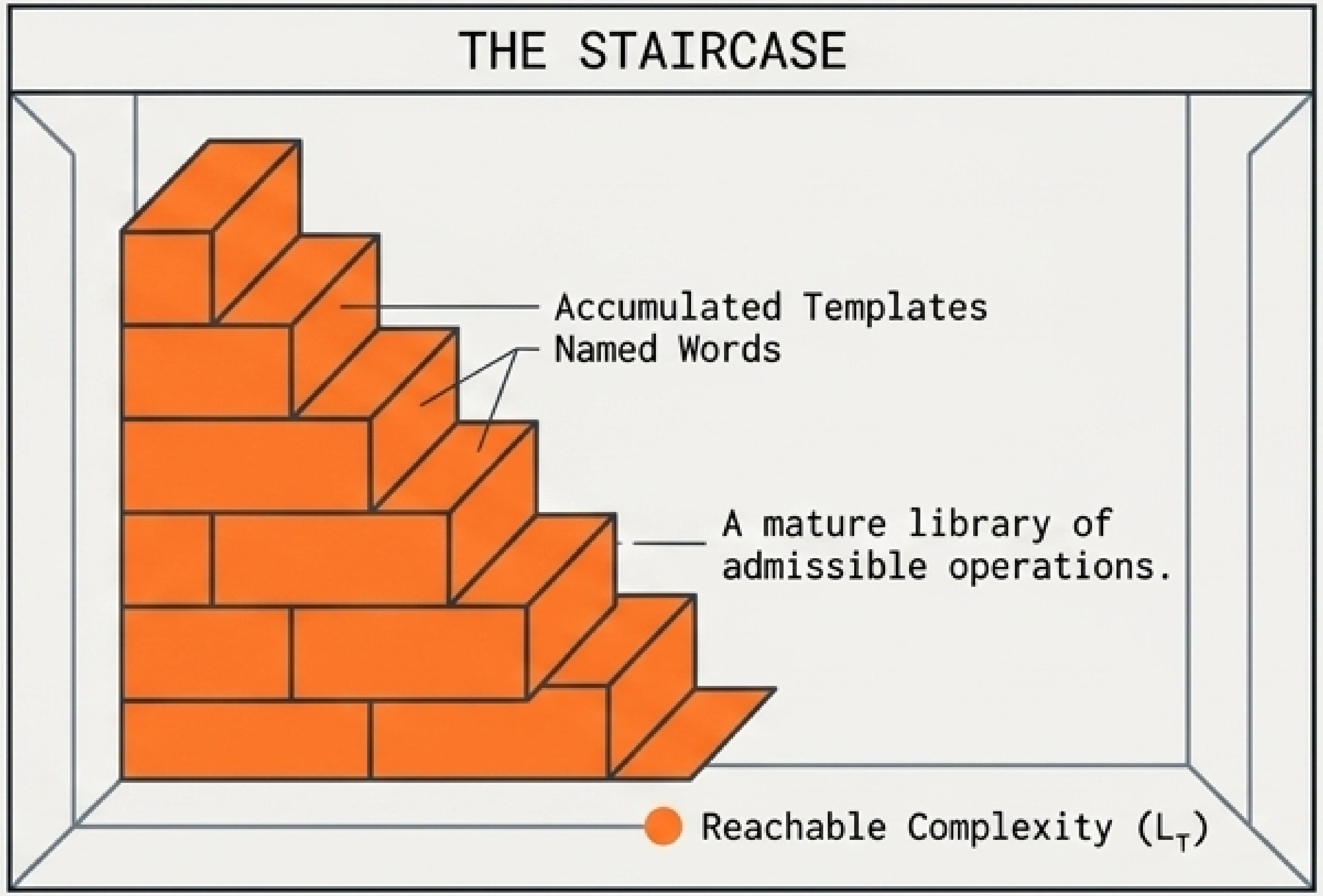
The Conclusion

An upper bound is not a two-sided estimate.

Using description length as a continuous, graded proxy for understanding extends the theorem far past what it actually supports.

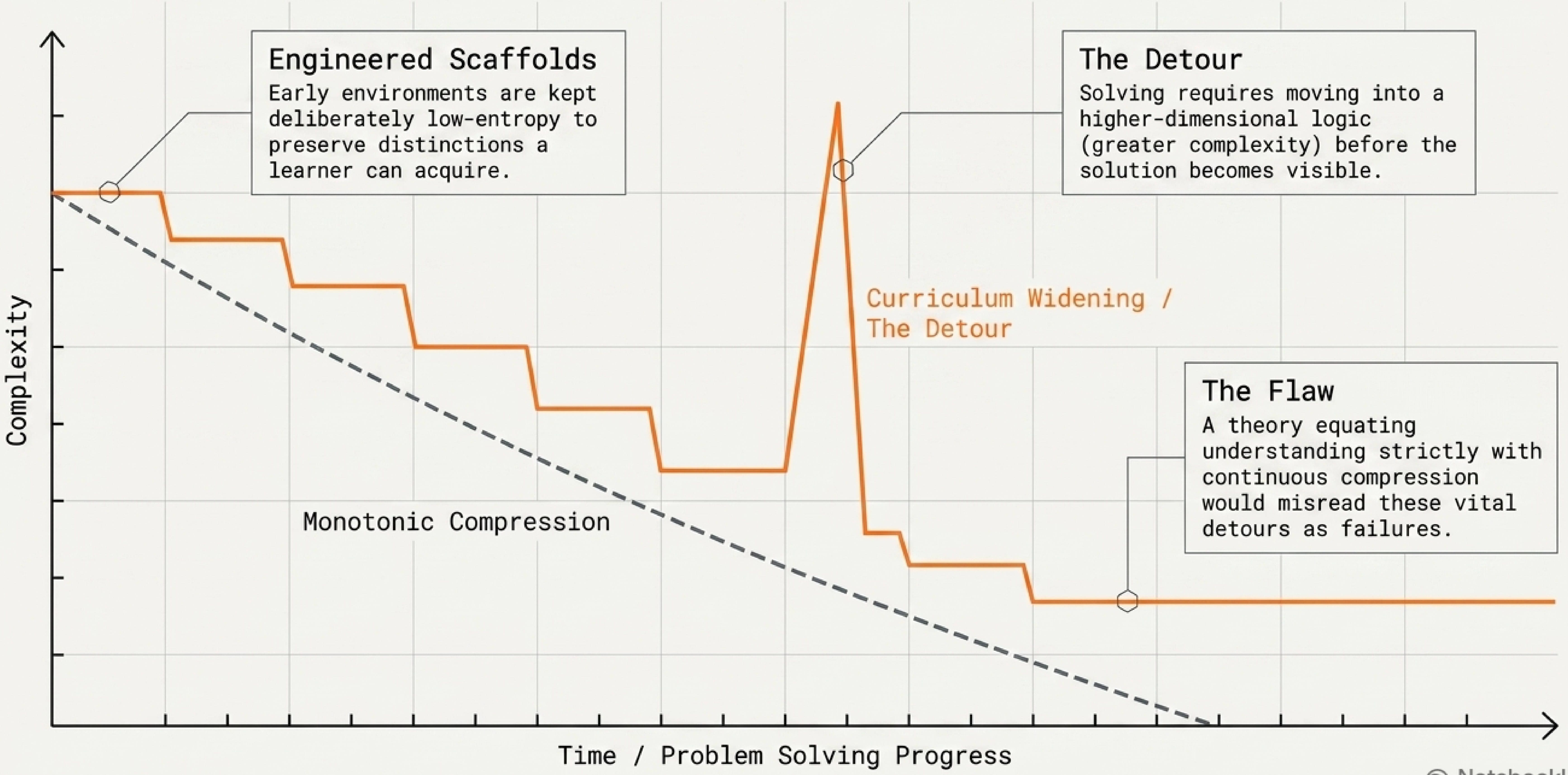
REACHABLE COMPLEXITY: THE REALITY OF MATURE SYSTEMS





- **The Forth Case:** Code is short not because it approaches the informational floor, but because it cheaply accesses an already-accumulated library of operations.
- **The ML Case:** Distributed neural representations routinely violate the “faithful encoding” assumption required by the Proxy Theorem (that every distinction costs at least one bit).
- **Takeaway:** Terseness reports the prior existence of an expanded library, not proximity to absolute truth.

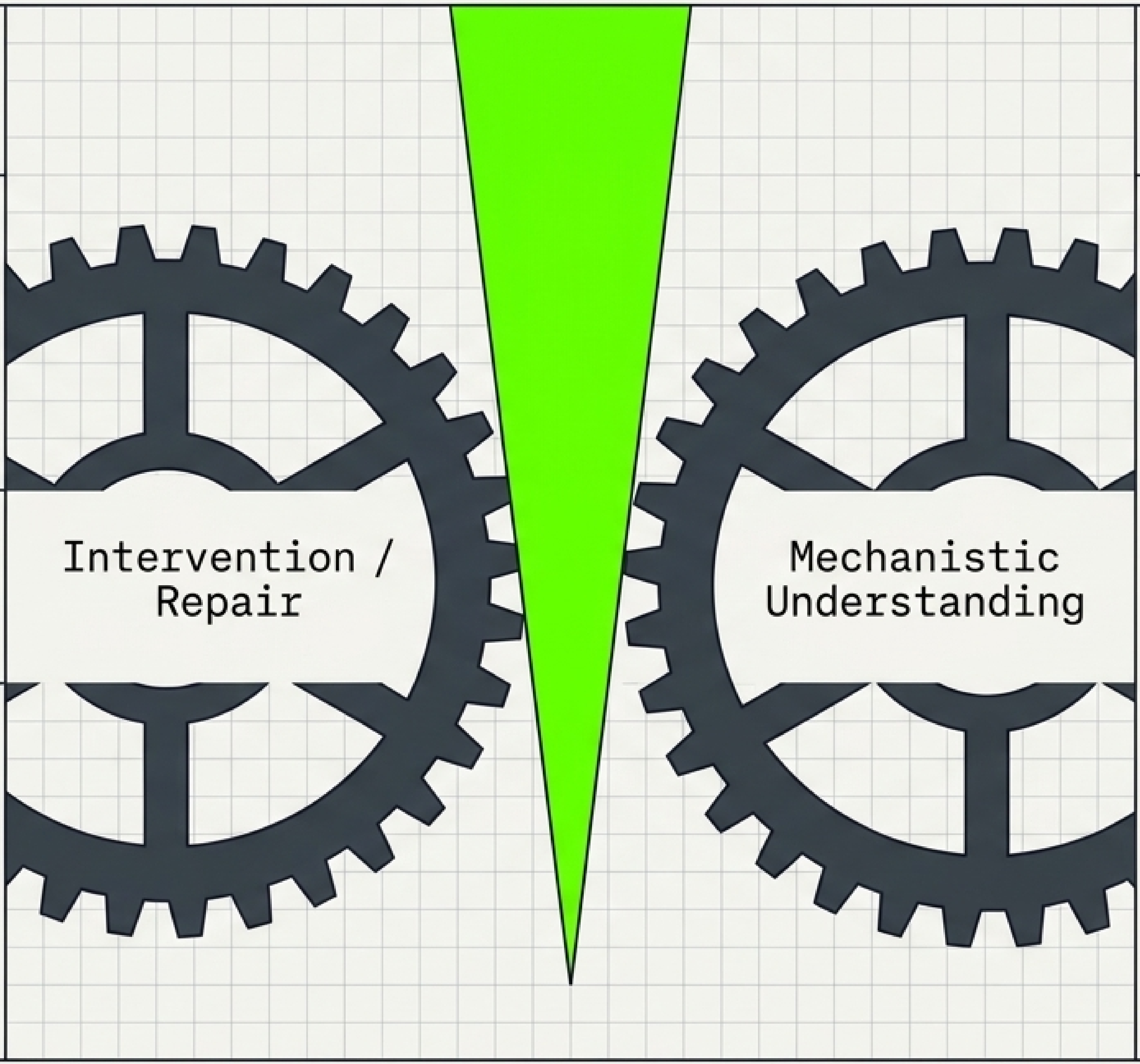
The Curriculum Detour: Why Complexity Precedes Simplicity



Restoration Before Explanation: A Structural Decoupling

The False Habit of Practice

The assumption that a system must be explicitly explained before it can be reliably repaired.



The Reality

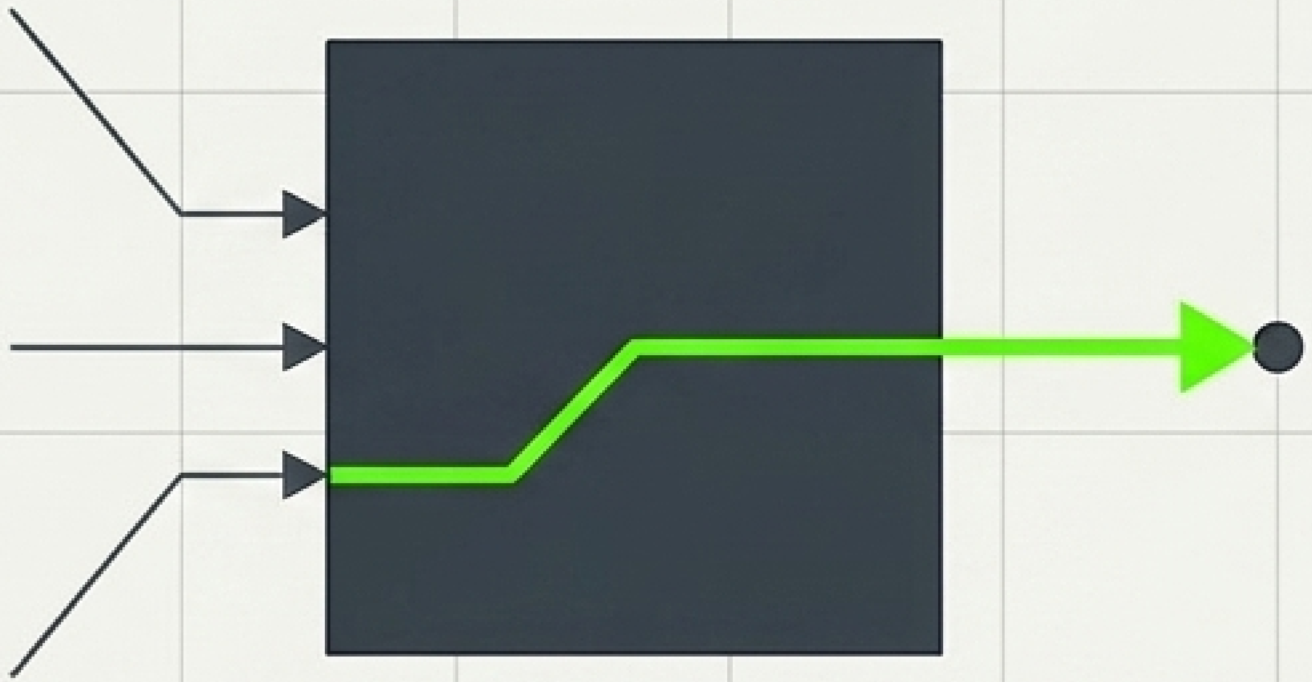
Restoration of competent function is not gated on prior explanation. It routinely precedes it, existing as a separate structural achievement.

The Convergent Truth

Two independent theories—recoverable persistence (explanation as distinction repair) and admissibility tracking (adequacy vs transparency)—prove this ordering is mathematically false.

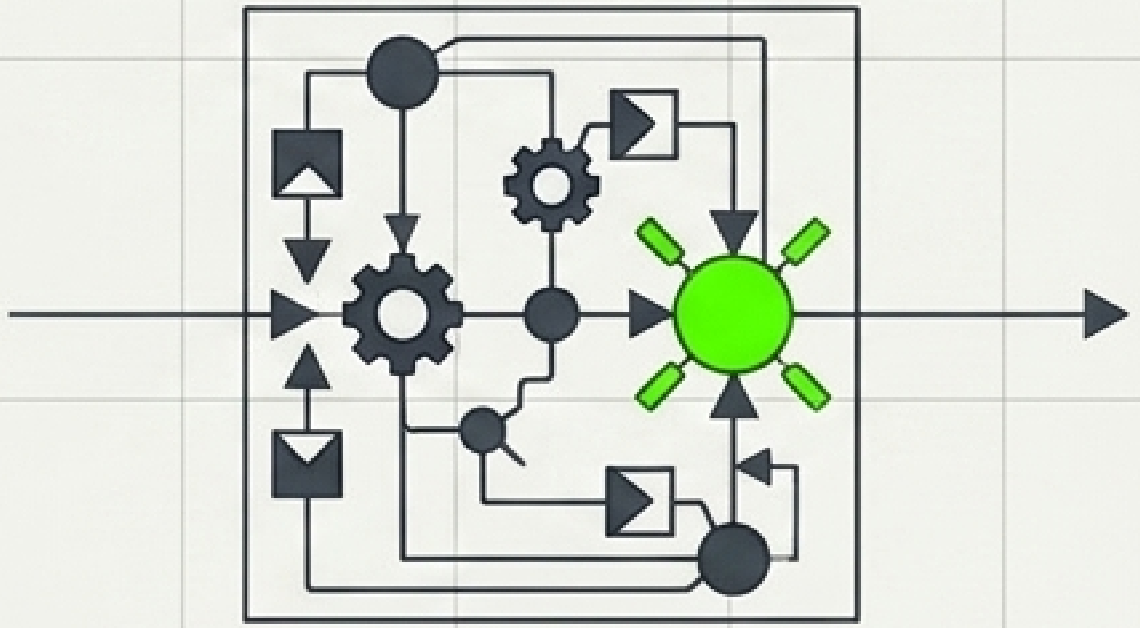
Functional Adequacy vs. Architectural Transparency

Functional Adequacy (Restoration)



The system's downstream behavior correctly tracks a distinction at better than chance under a controlled audit, regardless of internal mechanics.

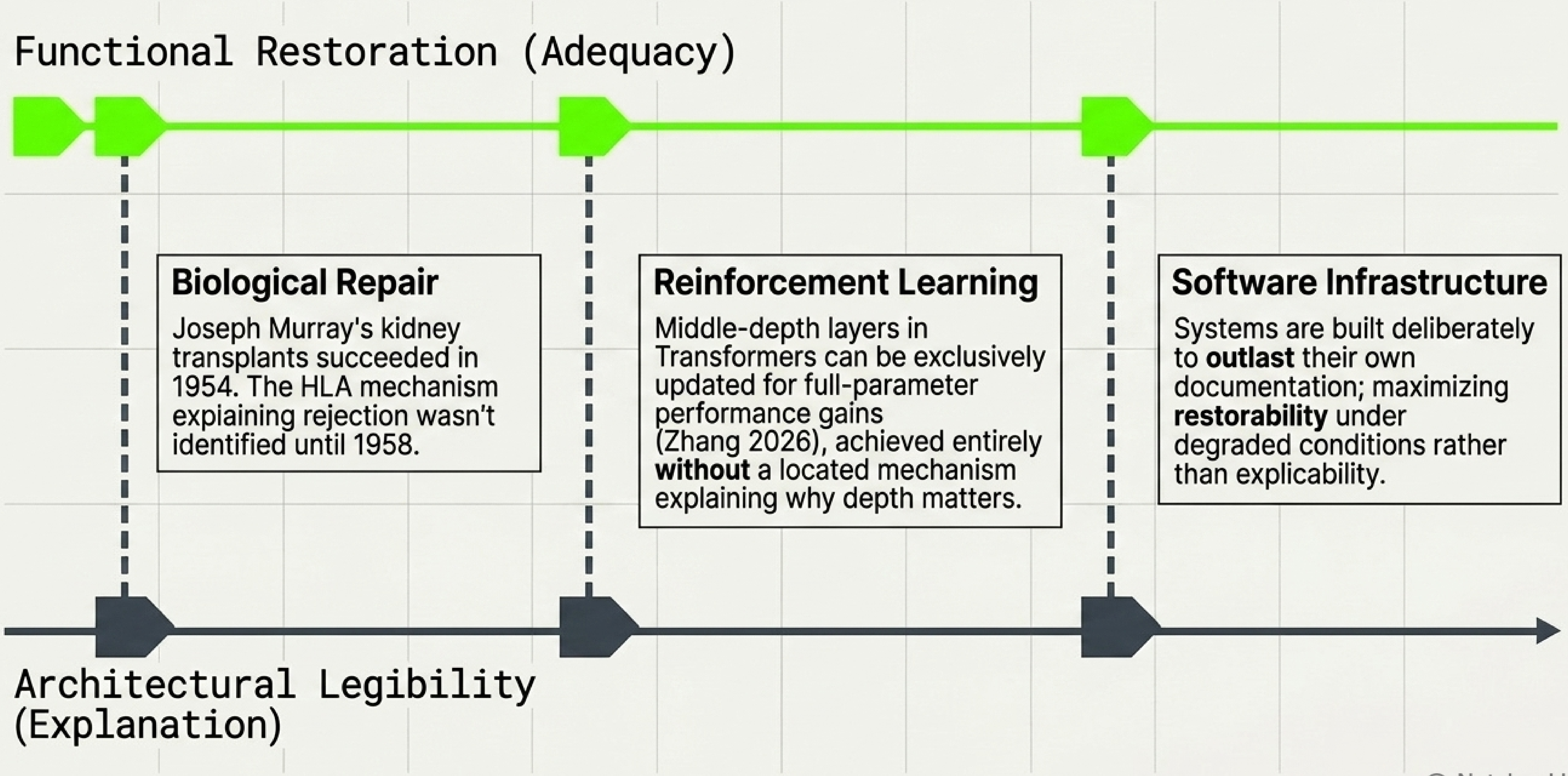
Architectural Transparency (Explanation)



The system maintains an explicit, inspectable internal object whose state can be checked against the distinction directly.

The Decoupling: A system can easily clear the first bar while remaining indefinitely short of the second.

Domain Evidence: The Temporal Gap



The Decoupling Framework: A Unified Meta-Pattern

The Unified Theme: Conflating a shared operational outcome with a shared internal mechanism.

Fallacy: Boundaries are geometric / passive.	Correction: Boundaries are categorical operations (Screening / Pursuit / Deletion).
Fallacy: Compression equals Understanding.	Correction: Description Length is formally **decoupled** from **Distinguishing Capacity.**
Fallacy: Explanation precedes Restoration.	Correction: Functional Adequacy is **structurally independent of Architectural Transparency.**

Measure the Right Variables

Rule 1: Stop treating boundaries as metaphors.

Identify whether you are measuring synchronic screening, stored pursuit, or diachronic deletion. They are not interchangeable.

Rule 2: Measure distinguishing capacity directly.

Stop using bit count or description length as a continuous proxy for understanding. Audit for superposition and faithful encoding.

Rule 3: Separate the audit from the explanation.

Accept that systems can be functionally adequate and successfully repaired without an explicit, inspectable internal architecture.

Closing Note: Precision in category isolation is the highest form of system evaluation. Avoid the flattening move.